Original Dataset Statistics

ORIGINAL TRAIN

📍 Target Label Distribution Balance:
- Binary Classes:
  - NOFF: 3562 (61.43%)
  - OFF: 2236 (38.57%)
- Multiclass Subcategories:
  - NO: 3562 (61.43%)
  - OO: 1950 (33.63%)
  - OR: 218 (3.76%)
  - OS: 68 (1.17%)

ORIGINAL TEST

📍 Target Label Distribution Balance:
- Binary Classes:
  - NOFF: 896 (61.79%)
  - OFF: 554 (38.21%)
- Multiclass Subcategories:
  - NO: 896 (61.79%)
  - OO: 486 (33.52%)
  - OR: 49 (3.38%)
  - OS: 19 (1.31%)

Final Dataset Statistics

FINAL EXPANDED TRAIN

📍 Target Label Distribution Balance:
- Binary Classes:
  - NOFF: 5000 (25.00%)
  - OFF: 15000 (75.00%)
- Multiclass Subcategories:
  - NO: 5000 (25.00%)
  - OO: 5000 (25.00%)
  - OR: 5000 (25.00%)
  - OS: 5000 (25.00%)

FINAL EXPANDED TEST

📍 Target Label Distribution Balance:
- Binary Classes:
  - NOFF: 5000 (25.00%)
  - OFF: 15000 (75.00%)
- Multiclass Subcategories:
  - NO: 5000 (25.00%)
  - OO: 5000 (25.00%)
  - OR: 5000 (25.00%)
  - OS: 5000 (25.00%)

DUE TO COMPUTATION limitations , all the generated dataset was not used to evaluate

Normalizing Dataset

Dataset required some level of formatting , different ikar , ukar generates additional computation which can be reduced by eliminating such variabilities , using a normalizing map can resolve this issue

```
NORMALIZATION_MAP = {
    "ि": "ी",
    "ु": "ू",
    "ॅ": "े",
    "श": "स",
    "ष": "स",
    "ृ": "रि",
    "ऋ": "रि",
    "ँ": "ं",
    "ई": "इ",
    "ऊ": "उ",
    "ै": "े",
    "ौ": "ो",
    "ण": "न",
}
```

Which as a result , it gives the result

Un-normalized	Normalized

{
"ID": "ab1f3af3-56b3-4165-8d74-3e9fa35c1da8",
"Comment": "तिमीलाई थाहा छ यो कस्तो गरिबीको दिखावा हो 🙈",
"Label_Binary": "OFF",
"Label_Multiclass": "OS"
}
	{ "ID": "ab1f3af3-56b3-4165-8d74-3e9fa35c1da8",
	"Comment": "तीमीलाइ थाहा छ यो कस्तो गरीबीको दीखाव हो 🙈",
	"Label_Binary": "OFF",
	"Label_Multiclass": "OS"
	}

Cleaning Process and Annotation Process on given dataset

Some labels were mislabeled and subjective and did not capture the actual intent of the sentence

Before	After
}, { "ID": "lyowdk", "Comment": "हैट सुड्डो पो रैछौ त 😂😂", "Label_Binary": "OFF", "Label_Multiclass": "OO" },	{ "ID": "lyowdk", "Comment": "हैट सुड्डो पो रैछौ त 😂😂", "Label_Binary": "NOFF", "Label_Multiclass": "NO" },
}, { "ID": "oloyqy", "Comment": "ra tero sister thamel ko besiy "Label_Binary": "OFF", "Label_Multiclass": "OO" },	{ "ID": "oloyqy", "Comment": "ra tero sister thamel ko besiya", "Label_Binary": "OFF", "Label_Multiclass": "OS" },

```
{
    "ID": "vunuau",
    "Comment": "chorin",
    "Label_Binary": "NOFF",
    "Label_Multiclass": "NO"
},
```

Such small words do not give context , these cannot be classified as hat and non hate speech , so such words were removed upon noticing.

Results with and without emoji

<table>
<tr><th>Model</th><th>With Emoji</th><th>Without Emoji</th></tr>
<tr><td>LR</td><td><pre>...
Training Logistic Regression...

==
Logistic Regression - Binary Classification
==
Accuracy: 0.7841

Classification Report:
               precision   recall f1-score   support

Non-Offensive   0.78   0.91   0.84   896
    Offensive   0.80   0.58   0.67   554

     accuracy   0.78   1450
    macro avg   0.79   0.74   0.76   1450
 weighted avg   0.79   0.78   0.78   1450
</pre></td><td><pre>...
Training Logistic Regression...

==
Logistic Regression - Binary Classification
==
Accuracy: 0.7814

Classification Report:
               precision   recall f1-score   support

Non-Offensive   0.78   0.91   0.84   896
    Offensive   0.79   0.58   0.67   554

     accuracy   0.78   1450
    macro avg   0.78   0.74   0.75   1450
 weighted avg   0.78   0.78   0.77   1450
</pre></td></tr>
<tr><td>SVM</td><td><pre>...
Training SVM...

==
SVM - Binary Classification
==
Accuracy: 0.7821

Classification Report:
               precision   recall f1-score   support

Non-Offensive   0.79   0.89   0.83   896
    Offensive   0.77   0.61   0.68   554

     accuracy   0.78   1450
    macro avg   0.78   0.75   0.76   1450
 weighted avg   0.78   0.78   0.78   1450
</pre></td><td><pre>...
Training SVM...

==
SVM - Binary Classification
==
Accuracy: 0.7807

Classification Report:
               precision   recall f1-score   support

Non-Offensive   0.79   0.88   0.83   896
    Offensive   0.76   0.62   0.68   554

     accuracy   0.78   1450
    macro avg   0.78   0.75   0.76   1450
 weighted avg   0.78   0.78   0.78   1450
</pre></td></tr>
<tr><td>Random Forest</td><td><pre>...
Training Random Forest...

==
Random Forest - Binary Classification
==
Accuracy: 0.8048

Classification Report:
               precision   recall f1-score   support

Non-Offensive   0.79   0.93   0.85   896
    Offensive   0.83   0.61   0.70   554

     accuracy   0.80   1450
    macro avg   0.81   0.77   0.78   1450
 weighted avg   0.81   0.80   0.80   1450
</pre></td><td><pre>...
Training Random Forest...

==
Random Forest - Binary Classification
==
Accuracy: 0.7966

Classification Report:
               precision   recall f1-score   support

Non-Offensive   0.79   0.91   0.85   896
    Offensive   0.81   0.61   0.70   554

     accuracy   0.80   1450
    macro avg   0.80   0.76   0.77   1450
 weighted avg   0.80   0.80   0.79   1450
</pre></td></tr>
</table>

Random Forest Multiclass

```
Training Multiclass Random Forest...

============================================================
Random Forest Multiclass - Multiclass Classification
============================================================
Accuracy: 0.7710

Classification Report:
              precision    recall  f1-score   support

          NO       0.78      0.93      0.85       896
          OO       0.76      0.57      0.65       486
          OR       0.62      0.16      0.26        49
          OS       0.00      0.00      0.00        19

    accuracy                           0.77      1450
   macro avg       0.54      0.42      0.44      1450
weighted avg       0.76      0.77      0.75      1450
```

```
Training Multiclass Random Forest...

============================================================
Random Forest Multiclass - Multiclass Classification
============================================================
Accuracy: 0.7779

Classification Report:
              precision    recall  f1-score   support

          NO       0.78      0.94      0.85       896
          OO       0.77      0.58      0.66       486
          OR       0.64      0.18      0.29        49
          OS       0.00      0.00      0.00        19

    accuracy                           0.78      1450
   macro avg       0.55      0.42      0.45      1450
weighted avg       0.76      0.78      0.76      1450
```

MBERT

```
Training mBERT Binary...
Epoch 1/2, Loss: 0.6180
Epoch 2/2, Loss: 0.4933

============================================================
mBERT Binary - Binary Classification
============================================================
Accuracy: 0.8076

Classification Report:
               precision    recall  f1-score   support

Non-Offensive       0.83      0.87      0.85       896
    Offensive       0.77      0.70      0.74       554

     accuracy                           0.81      1450
    macro avg       0.80      0.79      0.79      1450
 weighted avg       0.81      0.81      0.81      1450
```

```
============================================================
mBERT Binary - Binary Classification
============================================================
Accuracy: 0.7903

Classification Report:
               precision    recall  f1-score   support

Non-Offensive       0.85      0.80      0.82       896
    Offensive       0.70      0.78      0.74       554

     accuracy                           0.79      1450
    macro avg       0.78      0.79      0.78      1450
 weighted avg       0.80      0.79      0.79      1450
```

MBERT Multicast

```
Training mBERT Multiclass...
Epoch 1/2, Loss: 0.8125
Epoch 2/2, Loss: 0.6415

============================================================
mBERT Multiclass - Multiclass Classification
============================================================
Accuracy: 0.7717

Classification Report:
              precision    recall  f1-score   support

          NO       0.79      0.92      0.85       896
          OO       0.72      0.60      0.66       486
          OR       0.00      0.00      0.00        49
          OS       0.00      0.00      0.00        19

    accuracy                           0.77      1450
   macro avg       0.38      0.38      0.38      1450
weighted avg       0.73      0.77      0.75      1450
```

```
============================================================
mBERT Multiclass - Multiclass Classification
============================================================
Accuracy: 0.7717

Classification Report:
              precision    recall  f1-score   support

          NO       0.84      0.83      0.84       896
          OO       0.69      0.72      0.70       486
          OR       0.52      0.47      0.49        49
          OS       0.00      0.00      0.00        19

    accuracy                           0.77      1450
   macro avg       0.51      0.51      0.51      1450
weighted avg       0.76      0.77      0.77      1450
```

We obtained better result while conducting experiment with cleaning given data including emoji and normalizing , we obtained better result in MBERT compared to results of other methods, based on future aspects of expansion for hate speech on not only nepali but other languages too.

Based on result of ML model results , dataset with emoji gave acceptable results , hence we only used dataset with emoji for quantum VQC section after creating mbert embeddings , we got the following

Qubits :5
Epoch: 30
Circuit Depth: 6
Pre-layer dim : 128

Classical Learning Rate = 1e-4
Quantum Weight = 0.005

Test accuracy: 0.6903

	precision	recall	f1-score	support
0	0.78	0.70	0.74	896
1	0.58	0.68	0.63	554
accuracy			0.69	1450
macro avg	0.68	0.69	0.68	1450
weighted avg	0.70	0.69	0.69	1450